Google Cloud

Certificate resources and citations

Course 1 resources and citations

Module 1: Introduction to cloud computing

Resources

Helpful resources and tips

- Common problems with labs/lab troubleshooting
- Google Docs Help Center
- Google Sheets Help Center
- How to use Google Slides

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Module 2: Data analytics in the cloud

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Module 3: The data life cycle

Resources

Importance of data management

- Elements of a data management plan
- Writing a DMP step by step

[Supplemental] Store and use data with SQL

- SQL tutorial

Summary of the data lifecycle

- Data lifecycle management: Definition & 5 stages
- Introduction to the data lifecycle

Overview of data team roles

- How to structure your data analytics team
- Welcome to the data team handbook

Types of automation for the data lifecycle

- Options for controlling data lifecycles
- What is data analytics automation? (Plus Benefits)

Versioning and holds for accurate data

- Version control
- Version control - Research data management

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Summary of the data lifecycle

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Module 4: The role of a cloud data analyst

Resources

Data analyst job responsibilities

- A day in the life of a data analyst—What's It really like?
- What does a data analyst do? Exploring the day-to-day of this tech career

End-to-end data management

- 10 best practices for analytics success (including 3 you can't ignore)
- Reimagine data analytics for the era of AI

Types of data access management

- IAM: There and back again using resource hierarchies
- Tips on building a network security policy in Google Cloud

Communicate and collaborate with stakeholders

- How to create a stakeholder engagement plan
- Identifying and analyzing stakeholders and their interests

The data request process

- Managing data requests
- Responding to data requests

Google Cloud's data storage and management tools

- Go from prototype to production quickly

Uses of generative AI in data analytics

- Get started with generic search
- How AI is rewriting the rules of data analysis

Guide to Google Cloud

- Cloud Billing documentation
- Free cloud features and trial offer
- Google Cloud console
- Google Cloud console overview
- Why Google Cloud

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Course 2 resources and citations

Module 1: Introduction to data management and storage in the cloud

Resources

Helpful resources and tips

- Common problems with labs/lab troubleshooting
- Google Docs Help Center
- Google Sheets Help Center
- How to use Google Slides

Guide to BigQuery

- Enable the BigQuery sandbox
- Query a public dataset with the Google Cloud console

Citations

Data storage and connections

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Module 2: Key components of data organization

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Module 3: Steps to find data

Resources

How to find data using BigQuery

- BigQuery explained: How to query your data
- BigQuery public datasets

Knowledge Catalog's data lineage feature

- Data lineage

Introducing the Open Knowledge Format

- Introducing the Open Knowledge Format

How to use the Knowledge Catalog data lineage feature

- Use data lineage with Google Cloud systems

BigQuery sharing enables data sharing

- Introduction to BigQuery sharing

How to use BigQuery sharing

- Manage data exchanges

- Manage subscriptions

Overview of Knowledge Catalog

- Knowledge Catalog overview
- Google pitches Agentic Data Cloud to help enterprises turn data into context for AI agents

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How to find data using BigQuery

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Introduction to BigQuery sharing

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How to use BigQuery sharing

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How to search for data with BigQuery

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Module 4: Techniques to access data

Resources

Auto-detection of schemas in BigQuery

- Using schema auto-detection

Cloud-based machine learning can train predictive models

- Introduction to machine learning on Gemini Enterprise Agent Platform

Managed notebooks

- Gemini Enterprise Agent Platform Workbench Jupyter Notebook tutorials
- Welcome to Colab!

Benefits of data partitioning

- Database partitions: How they work

Data partitioning reduces cloud costs

- Cloud Storage pricing
- Understanding Data Partitioning in Data Engineering: A Key to Scalability and Performance

Create a partitioned table

- Creating partitioned tables

Tips for interacting with partitioned tables

- Load data into partitioned tables
- Managing partitioned tables

How to manage Managed Service for Apache Spark clusters

- Create and manage Labels
- Stop and start Managed Service for Apache Spark clusters

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Module 1: Introduction to data transformation in the cloud

Resources

Helpful resources and tips

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- Google Docs Help Center
- Google Sheets Help Center
- How to use Google Slides

The data journey is a flexible path

- 7 data collection methods in business analytics

Impacts of the data journey on decision making

- Business analytics in action: Real-world examples and use cases

Techniques Used to Transform Data, Part 2

- Importance of proper formatting in data collection

Storage options for transformed data

- BigQuery overview

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Module 2: Handle raw data with data pipelines

Resources

Overview of data pipeline stages and benefits

- Data Pipelines: An Overview
- What Data Pipeline Architecture should I use?

Strategies and tools for efficient data ingestion

- Big data ingestion

Data transformation steps to ensure reliable data

- What are the most common data cleaning techniques?

Transformation approaches for securing data

- FBI Tech Tuesday: Protecting Against Personally Identifiable Information (PII) Theft
- How to safeguard personally identifiable information

Data dictionaries and monitoring help maintain data integrity

- Definitive Guide to Data Monitoring
- What Is a Data Dictionary?

Analysis and visualization after the data pipeline

- How to Meet WCAG (Quick Reference)
- Introduction to SQL

- Top data visualization tools for business professionals

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Overview of data pipeline stages and benefits

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Course 4 resources and citations

Module 1: Introduction to the power of storytelling: How to visualize data in the cloud

Resources

Helpful resources and tips

- Common problems with labs/lab troubleshooting
- Google Docs Help Center
- Google Sheets Help Center
- How to use Google Slides

Why visualization matters

- Information is beautiful
- Making events accessible
- Why data visualization is important
- Why is data visualization important? What is important in data visualization?

Foundational concepts of design: Structure and aesthetics

- Color choices in data visualizations: Importance and how to
- Visual hierarchy, Gutenberg diagram, F & Z pattern

Foundational concepts of design: Usability and accessibility

- Accessible data visualizations
- Data visualizations, charts, and graphs

Citations

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Module 2: Methods for visualization planning and design

Resources

Visualization techniques for different data types

- Data visualization resources: Types of charts and graphs for data viz
- How to choose the best chart or graph for your data

Choose an effective visualization

- Explore new ways to visualize your data

[Supplemental] Tips for effective teamwork

- Teams start with human connections

The importance of the user journey

- Web Content Accessibility guidelines

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Module 3: Access, explore, and report on data in the cloud

Resources

Cloud-based tools for accessing and visualizing data

- Looker for business intelligence
- Data Studio

Actionable insights through data exploration

- Unlock the power of data analytics with Looker: Rethinking business intelligence

Customize dimensions and measures in Data Studio

- About the "How to" guide

Case study: Benefits of using data reports

- 11 data visualization techniques for every use-case with examples
- 17 data visualization techniques all professionals should know
- Power BI dashboards vs reports: A comprehensive guide

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Guide to Data Studio

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Module 4: Enterprise business analytics

Resources

Data governance for safeguarding data

- Best practices In data governance
- What is data governance?

Insights from a single dataset

- What is exploratory data analysis?

Guide to Looker Enterprise

- Create and edit dashboards and reports
- How Looker generates SQL
- Looker introduction

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Introduction to enterprise-grade visualization tools

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Live dashboard features

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Guide to Looker Enterprise

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The changing role of a cloud data analyst

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Module 5: Explore the developer environment

Resources

Guide to LookML

- Introduction to LookML
- LookML data tests: Recommendations and best practices
- LookML terms and concepts
- Types of files in a LookML project
- Using version control and deploying

Explore a dashboard file's LookML code

- Dashboard element list
- LookML dashboard parameters

Citations

Bring data tools together with an IDE

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The elements of LookML

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The benefits of using version control

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Explore a dashboard file's LookML code

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Improve performance with caching

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Course 5 resources and citations

Module 1: Introduction: Prepare for a cloud data analyst job

Resources

Tools and tips for resume preparation

- Sample Resume Rough Draft
- Sample Resume Final Draft
- Google Docs
- How to Make a Resume: 2023 Resume Writing Guide
- Résumé Workshop

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The cloud data analytics hiring process

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Share your project work with a portfolio

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GenAI's role in skill development

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Highlight new skills in your resume

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Tools and tips for resume preparation

- Coursera Staff. (2023, November 29). *How to make a resume: 2023 resume writing guide*. Coursera.
- Purdue Online Writing Lab. (n.d.). *Résumé workshop*. Purdue University.

Prepare for interviews

- Carleton, S. C. (2020, March 19). Top 4 analytics interview questions and how to prepare. *Graduate Blog*.
- Columbia University Center for Career Education. (n.d.). *Prepare for a software or technical interview*.
- Marquette University. (n.d.). *Interviewing guide*.

Module 2: The capstone project

Resources

Incorporate your projects into your portfolio

- Creating an account on GitHub
- How to use Google Sites
- Welcome to Tableau Public
- When, why & how—to get started with Kaggle?

Citations

The value and benefits of the capstone project

- Singh, R. K. (2023, June 3). *Advantage of capstone project*. Linkedin.

Capstone project wrap-up

- Singh, R. K. (2023, June 3). *Advantage of capstone project*. Linkedin.

Module 3: Put your Cloud Data Analytics Certificate to work

Resources

Showcase your certificate

- Big Interview
- Create a resume in Google Docs
- Linkedin: Add and remove skills on your profile
- Linkedin: Edit the About section on your profile
- Linkedin: Manage licenses & certifications
- Search resume examples
- Start a resume

Claim your Google Cloud Data Analytics Certificate badge!

- How can I add my badge to my LinkedIn profile and share to my feed?
- How to add a badge to Linkedin?
- Welcome to the Credly Help Center